THE COKERNEL OF CYCLES UNDER PERFECT CLOSURE IS P-POWER TORSION

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Allowed standard prerequisites (editorial).

- f-power torsion modules (More on Algebra, Definition more-algebra-definition-f-power-torsion, canonical tag 05E6). cited at statement lines 368-369. Defines the p-power torsion notion the cokernel is claimed to have.
- Multiplicities of cycles under field extension (relative-cycles chapter, Lemma lemma-multiplicities-field-extension, canonical tag 0H4F). cited at proof lines 378-379. Computes the multiplicities appearing in the base change of cycles along field extensions, upgrading torsion to p-power torsion.

Source reference. Original definition 05E6: sources/stacks-chow/tags.txt, source lines 6979–6979. Cited at statement lines 368-369: defines p-power torsion for the cokernel.

Source reference. Original lemma 0H4F: sources/stacks-chow/tags.txt, source lines 21350–21350. Cited at proof lines 378-379: p-power torsion of the cokernel elements.

Lemma 0.2. *Let k be a field of characteristic $p > 0$ with perfect closure k^{perf}. Let X be an algebraic scheme over k. Let $r \geq 0$ be an integer. The cokernel of the injective map $Z_r(X) \to Z_r(X_{k^{perf}})$ is a p-power torsion module (More on Algebra, Definition 05E6).*

Proof. Since X is quasi-compact, the abelian group $Z_r(X)$ is free with basis given by the integral closed subschemes of dimension r. Similarly for $Z_r(X_{k^{perf}})$. Since $X_{k^{perf}} \to X$ is a homeomorphism, it follows that $Z_r(X) \to Z_r(X_{k^{perf}})$ is injective with torsion cokernel. Every element in the cokernel is p-power torsion by Lemma 0H4F. □

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